

Class 10 Matrix Operations

An $m \times n$ matrix consists of _____ column vectors with _____ entries. So there are _____ rows and _____ columns.

Notation

a_{ij} is

The matrix A is also denoted as $A =$
or as $A =$

4 operations on matrices

1. Matrix Addition
2. Scalar Multiplication
3. Matrix Multiplication

Example $A = \begin{bmatrix} 2 & 0 & 1 \\ -2 & 1 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 2 \\ -2 & 4 \\ 0 & 3 \end{bmatrix}$

$$3A =$$

$$A + B =$$

$$AB =$$

$$BA =$$

Operation #4. Transpose of a matrix

If A is any $m \times n$ matrix, the transpose of A is

Continue example above

$$A^T =$$

Key Idea: Connection between matrix multiplication and composition of linear transformations.

Explore:

Let $S: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ so that $S(x, y) = (y, x, x - y)$

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ so that $T(x, y, z) = (2x, 3y, x + y + z)$

1. What is $T \circ S$?

2. What is the matrix A for T ?

$A =$

3. Matrix B for S is

4. Matrix C for $T \circ S$ is

5. Compute the matrix product BA